You are required to finish part 1 to part 5 below.

Part 1. Select and secure access to cloud environment

1.1. Discuss and compare different cloud computing solutions, models and services according to business requirements and needs.

For each business requirement / need, two possible solutions are given, one is better than the other. You are required to discuss and compare the two solutions for each requirement, also explain which one is better.

Requirement #1: The client needs a cloud solution for hosting a database that can handle large amounts of unstructured data and can scale rapidly to accommodate unpredictable usage spikes.

Solution A: Using RDS (Relational Database Service)

Solution B: Using DynamoDB

Your answer:

Comparation:

RDS have a relational database management system which can handle structured data well . the scalability managed manually and the performance is high for smaller data. RDS is usually a bit more costly than other solutions.

DynamoDB is NoSQL document database with a semi-structured or unstructured data which can be scaled automatically based on the usage spikes. This solution is usually cheaper than the RDS.

Which solution is better?

DynamoDB is a better solution for the client because its ability to deal with unstructured data and scale rapidly to accommodate unpredictable usage spikes as the customer required.

References:

https://www.projectpro.io/article/amazon-rds-vs-dynamodb/810

Requirement #2: The client needs a cloud storage solution for storing publicly accessible static assets (images, videos, static HTML/JS/CSS files) for their website.

Solution A: Using EC2

Solution B: Using S3

Your answer:

Comparation:

EC2 is a virtual server in the cloud that gives users as many instances as they may need, based on the operating systems they prefer, without the need of a physical server/computer. It is storing operating systems and files which related to the application.

S3 is a storage service without the need of a physical devices for storage. It gives the ability to store data and enjoy the scalability it offers. The data is available at any time and easily accessible.

Which solution is better?

S3 is the solution when looking for a storage solution for storing publicly accessible static assets. It gives the ability to store data objects as the clients required (images and videos for instance) with high availability at any time.

References:

https://www.whizlabs.com/blog/amazon-ec2-vs-amazon-s3-comparison-guide/#Amazon_EC2_vs_Amazon_S3_What's_the_difference

Requirement #3: The client needs a cloud solution to run a web application that can handle high traffic volumes during peak hours in the daytime.

Solution A: Using a powerful EC2 instance.

Solution B: Using an Autoscaling group with a load balancer and multiple less powerful EC2 instances.

Your answer:

Comparation:

One powerful EC2 instance has high CPU and memory capacities that could work well with high-performance workloads and handle high traffic, but it is still have limitations with the power and memory so it can get overwhelmed at some point during peak hours.

Using autoscaling group with a load balancer could dynamically adding more and more instances depending on the volume of traffic during peak hours and shut them when there is a decrease in traffic volume. That way there is no need to use the most powerful EC2 instance which can cost more money as well.

Which solution is better?

Autoscaling group with a load balancer is a better solution in the case, because the traffic is increasing or decreasing consistently depending on the peak hours.

References:

https://avinetworks.com/glossary/auto-scaling/

1.2. Identify user access level / protocol and create user.

You are required to develop a cloud solution for hosting a web application on AWS platform. The web application includes the following components

- A backend python app
- A SQL database
- Multiple static files.

Before getting started with the development, you need to

- Identify user access protocols and policies according to development requirements
- Configure required AWS IAM policies with required permissions
- Create AWS IAM groups and assign IAM policies the groups
- Create and assign users to IAM groups

A. Identify user access protocols and policies according to business requirements and needs.

You need to explain what are the different users to be created and access level to each type of user. E.g. a developer type of user, with full access to EC2 / RDS / S3 services.

Your answer:

User 1 – S3-Support with read-only access to the S3 service.

User 2 – EC2-Support with read-only access to the EC2 service.

User 3 – EC2-Admin- with inline policy which allows the user to view and start/stop EC2 instances.

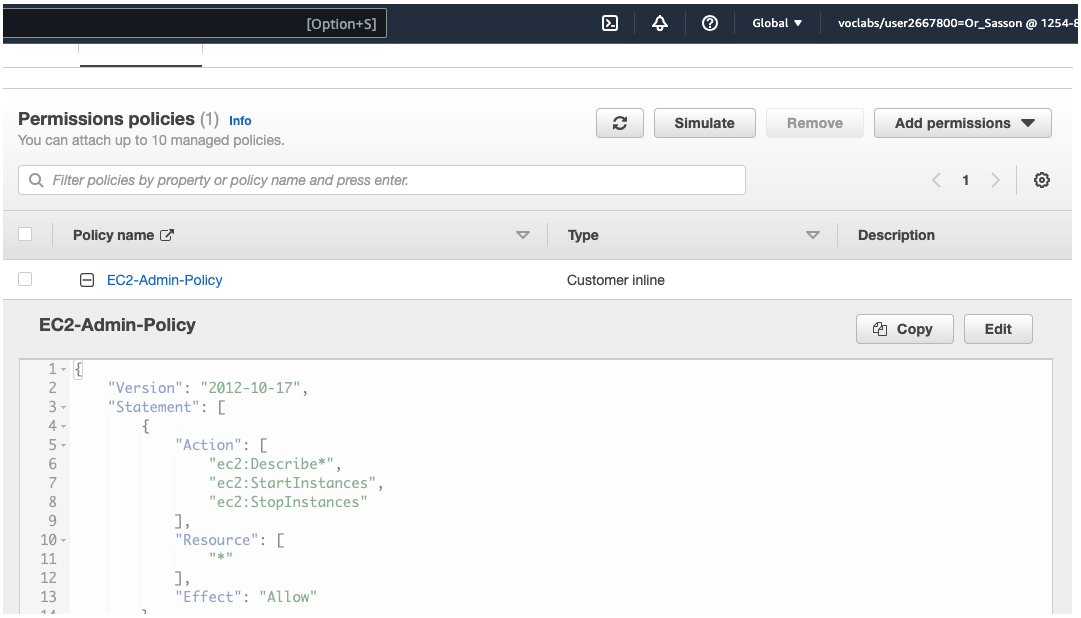
B. Configure required AWS IAM policies with required permissions

You need to logon to AWS cloud platform and create IAM policies according to user access protocols and policies identified above.

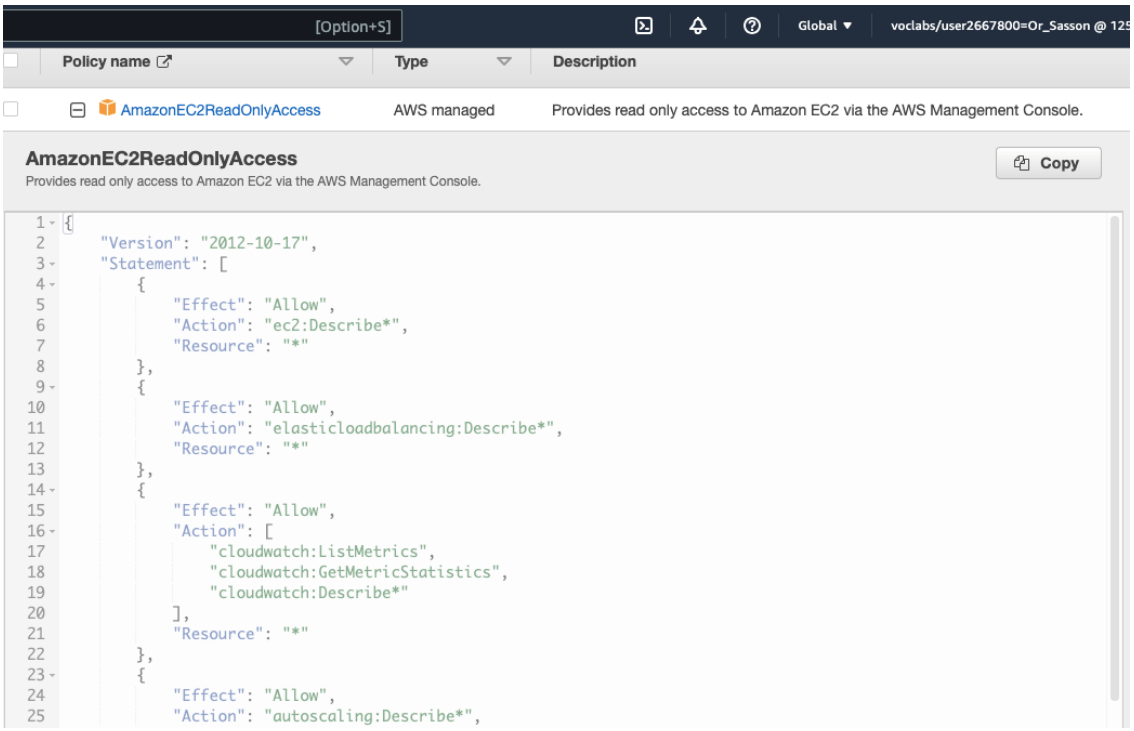
Your answer:

Provide screenshot of all the IAM policies created and explain the policy with one or two sentences.

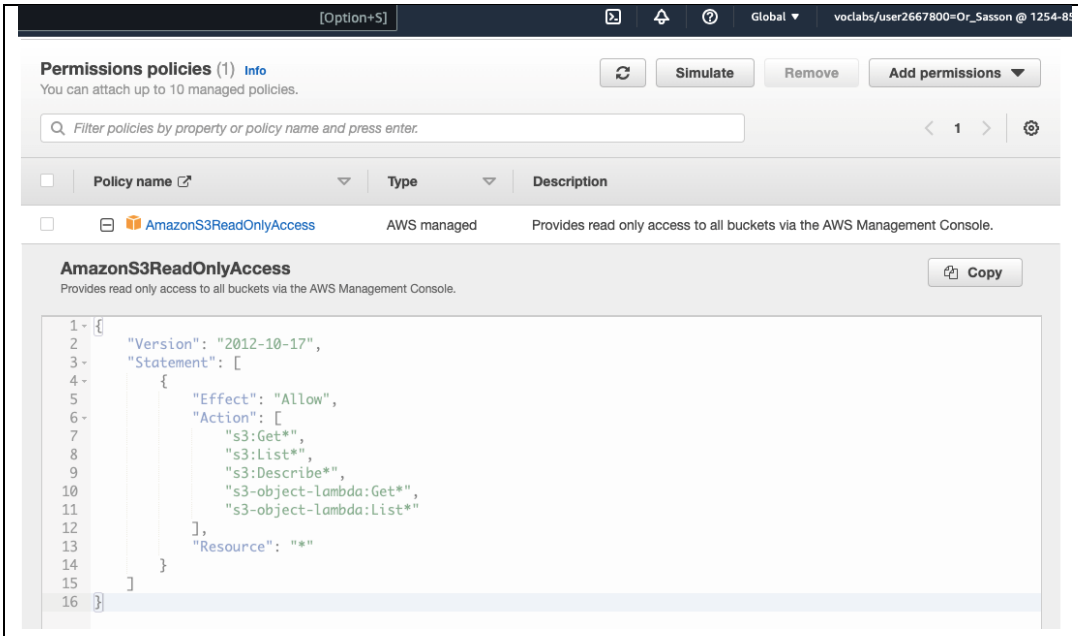
This policy is usually assigned to one user or group only and allow them to start or stop EC2 instances.



This policy allows the user to view the information of the EC2 instances without the ability to modify them.



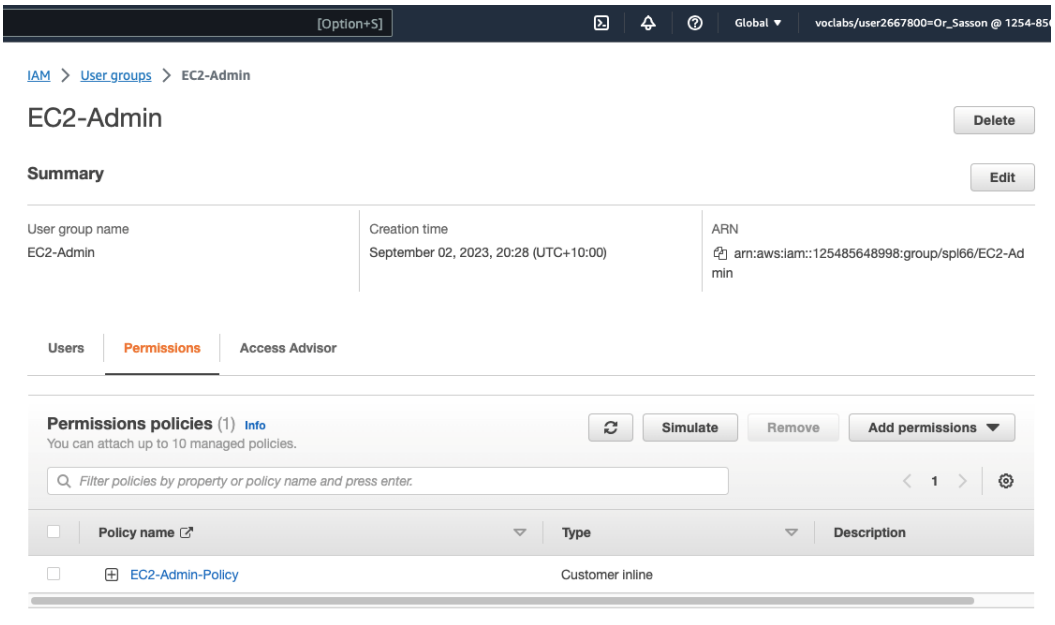
This policy allows the user to view the information of the S3 without to modify it.

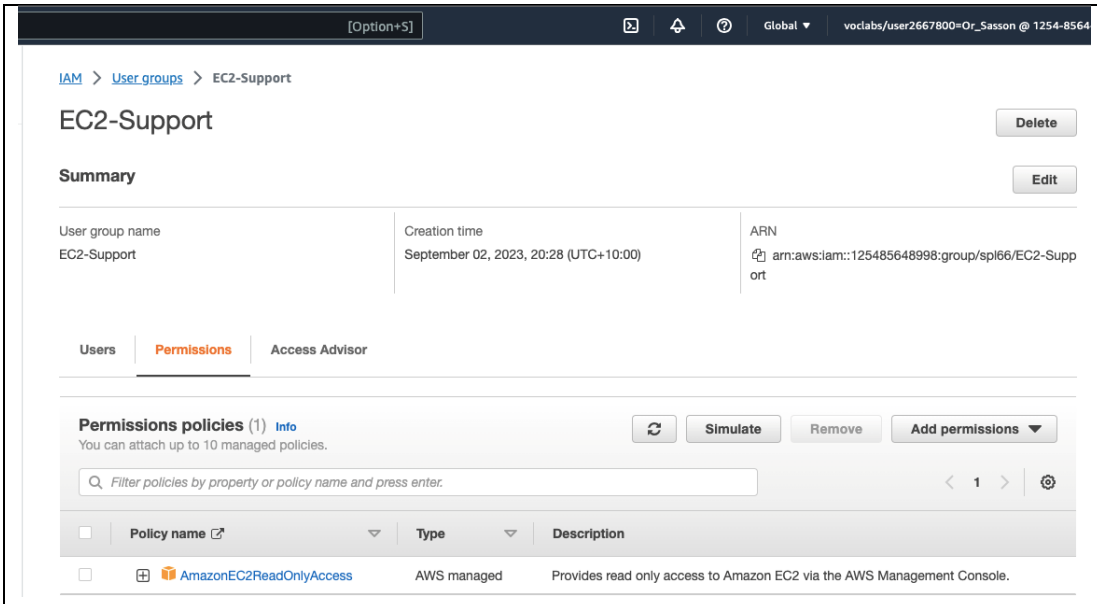


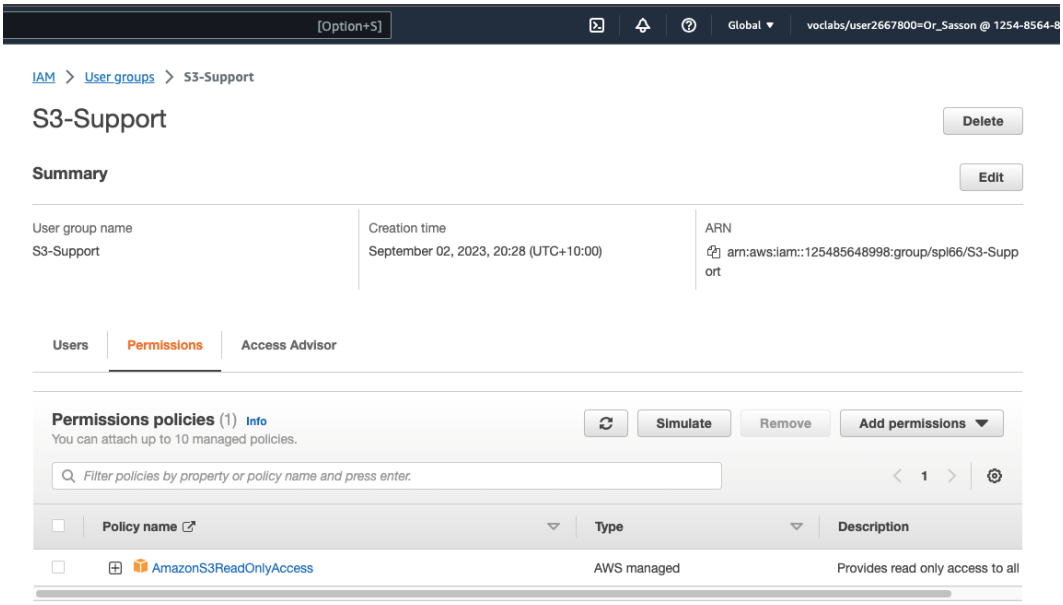
C. Create AWS IAM groups and assign IAM policies the groups

Your answer:

Provide screenshot of all the IAM groups created. In the screenshot, you must show that require IAM policies has been assigned to each IAM group.



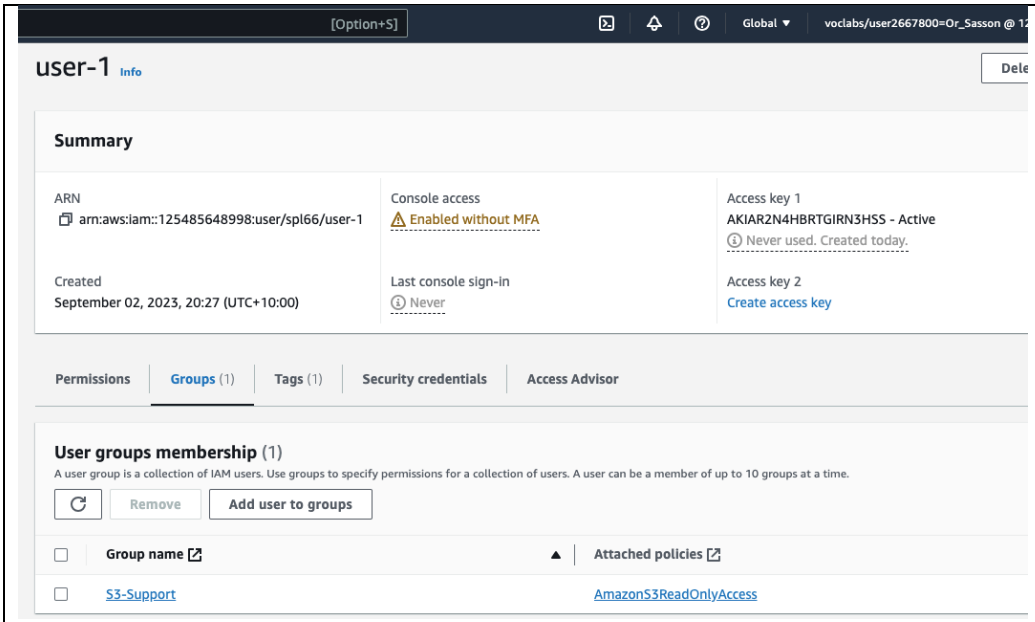


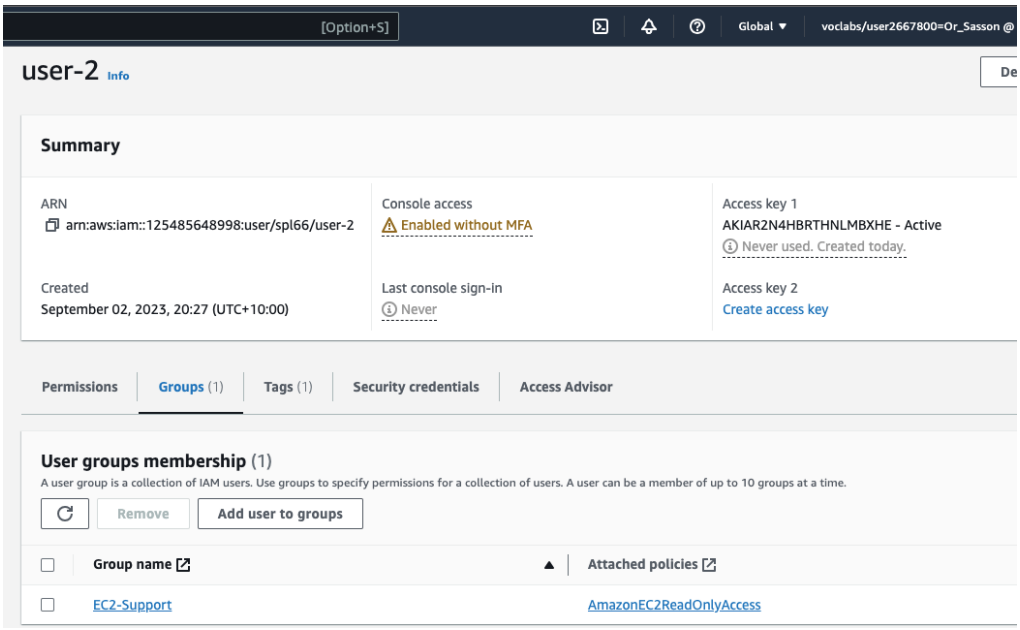


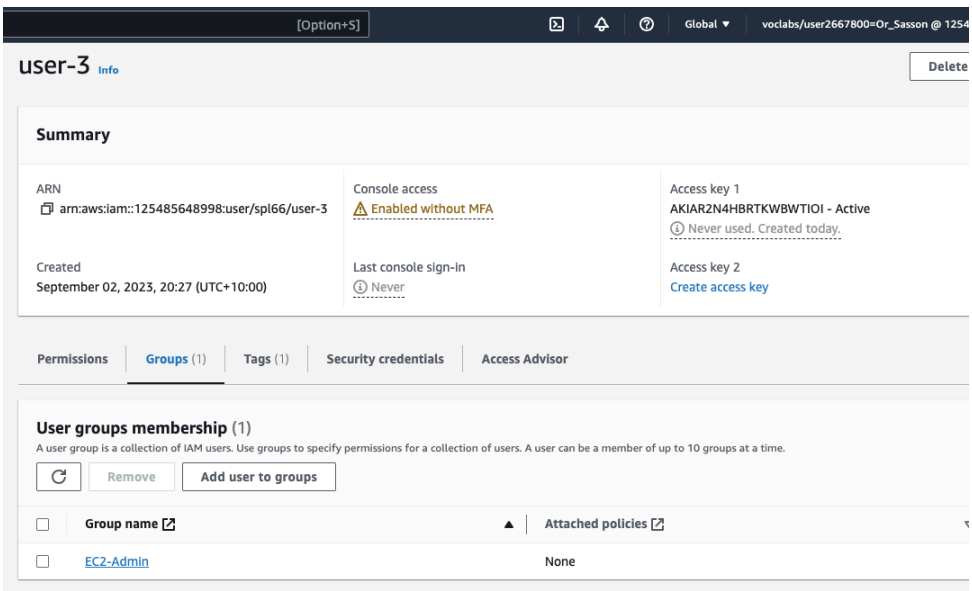
D. Create and assign users to IAM groups

Your answer:

Provide screenshot of all the IAM users created. In the screenshot, you must show that the users have been assigned to the correct group.


[Option+S]
Global
user-1 Info
Summary
ARN
arn:aws:iam::125485648998:user/spl66/user-1
Console access
Enabled without MFA
Access key 1
AKIAR2N4HBRTGIRN3HSS - Active
Never used. Created today.
Created
September 02, 2023, 20:27 (UTC+10:00)
Last console sign-in
Never
Access key 2
Create access key
Permissions
Groups (1)
Tags (1)
Security credentials
Access Advisor
User groups membership (1)
A user group is a collection of IAM users. Use groups to specify permissions for a collection of users. A user can be a member of up to 10 groups at a time.
Remove
Add user to groups
Group name
Attached policies
S3-Support
AmazonS3ReadOnlyAccess


[Option+S]
Global
user-2 Info
Summary
ARN
arn:aws:iam::125485648998:user/spl66/user-2
Console access
Enabled without MFA
Access key 1
AKIAR2N4HBRTHNLMBXHE - Active
Never used. Created today.
Created
September 02, 2023, 20:27 (UTC+10:00)
Last console sign-in
Never
Access key 2
Create access key
Permissions
Groups (1)
Tags (1)
Security credentials
Access Advisor
User groups membership (1)
A user group is a collection of IAM users. Use groups to specify permissions for a collection of users. A user can be a member of up to 10 groups at a time.
Remove
Add user to groups
Group name
Attached policies
EC2-Support
AmazonEC2ReadOnlyAccess


[Option+S]
Global
user-3 Info
Delete
Summary
ARN
arn:aws:iam::125485648998:user/spl66/user-3
Console access
Enabled without MFA
Access key 1
AKIAR2N4HBRTKWBWTIOI - Active
Never used. Created today.
Created
September 02, 2023, 20:27 (UTC+10:00)
Last console sign-in
Never
Access key 2
Create access key
Permissions
Groups (1)
Tags (1)
Security credentials
Access Advisor
User groups membership (1)
A user group is a collection of IAM users. Use groups to specify permissions for a collection of users. A user can be a member of up to 10 groups at a time.
Remove
Add user to groups
Group name
Attached policies
EC2-Admin
None

Part 2. Create and configure an Amazon Virtual Private Cloud (Amazon VPC) network instance

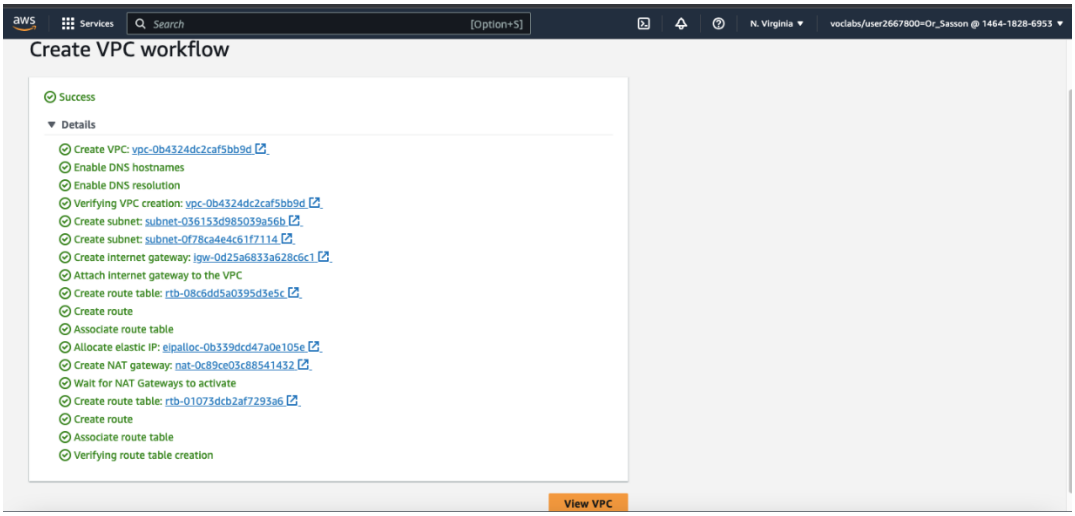
In this part, you are required to create an Amazon VPC instance with the following requirements

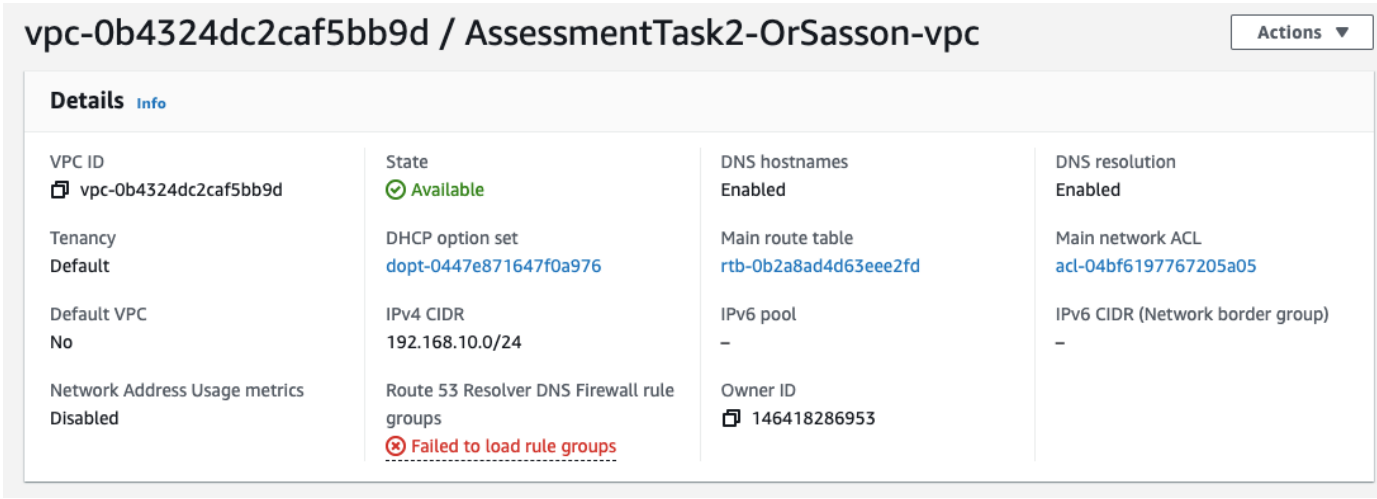
- CIDR 192.168.10.0/24
- Two subnets
  Subnet 1: 192.168.10.0/25 (network address: 192.168.10.0, broadcast address: 192.168.10.127)
  Subnet 2: 192.168.10.128/25 (network address: 192.168.10.128, broadcast address: 192.168.10.255)
- An Internet gateway allow both subnets to connect to the Internet

2.1. Create the required VPC and associated resources.

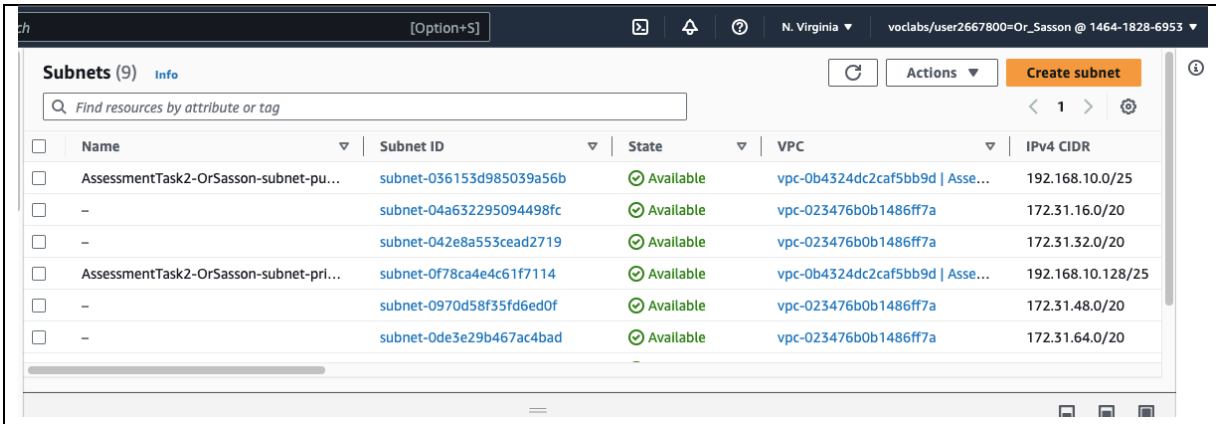
Your answer:

Provide screenshot of VPC instance created

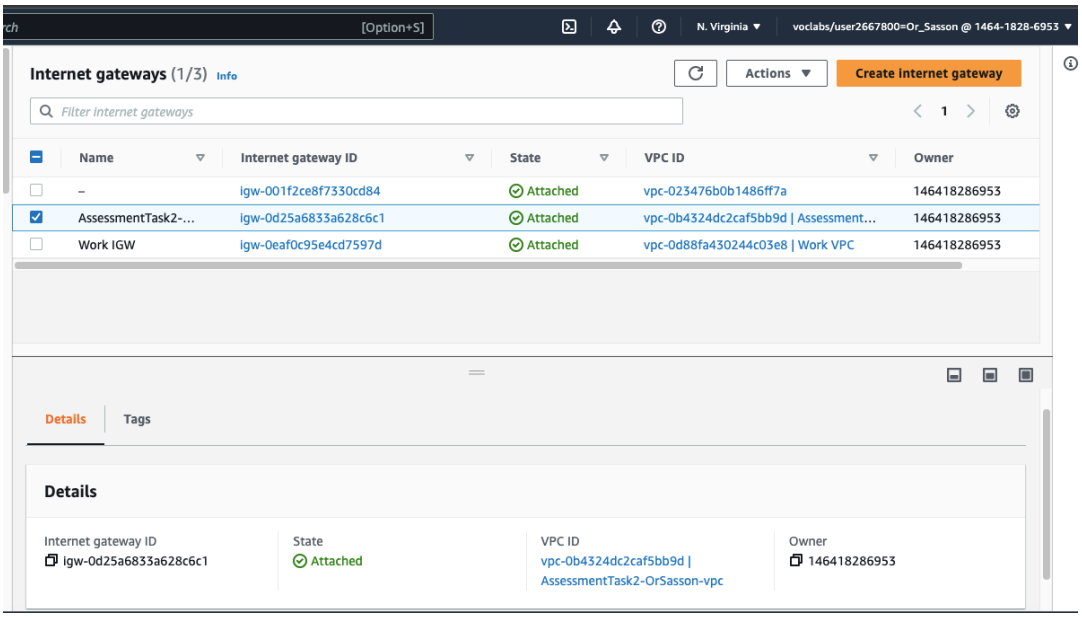




Provide screenshot of the two subnets instance created

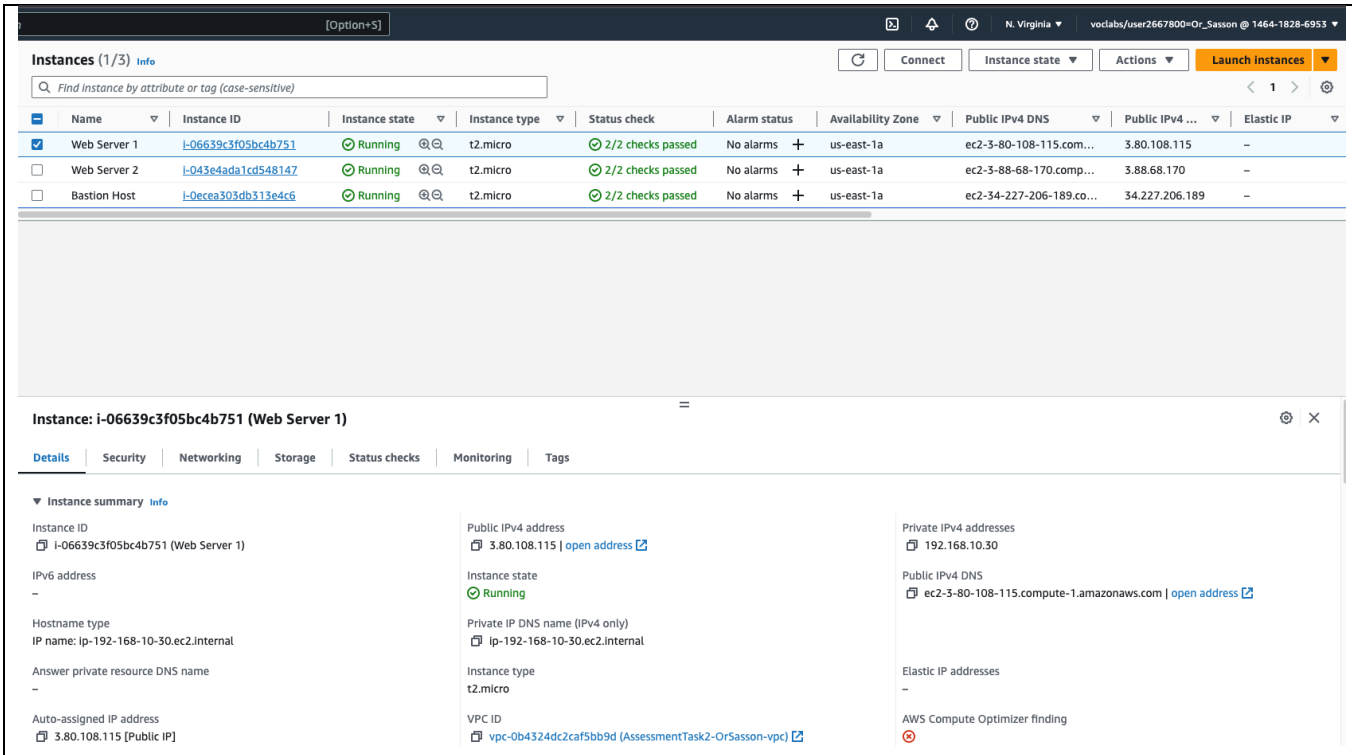


Provide screenshot of the Internet gateway instance created



2.2. Create two EC2 instances to verify the VPC create above

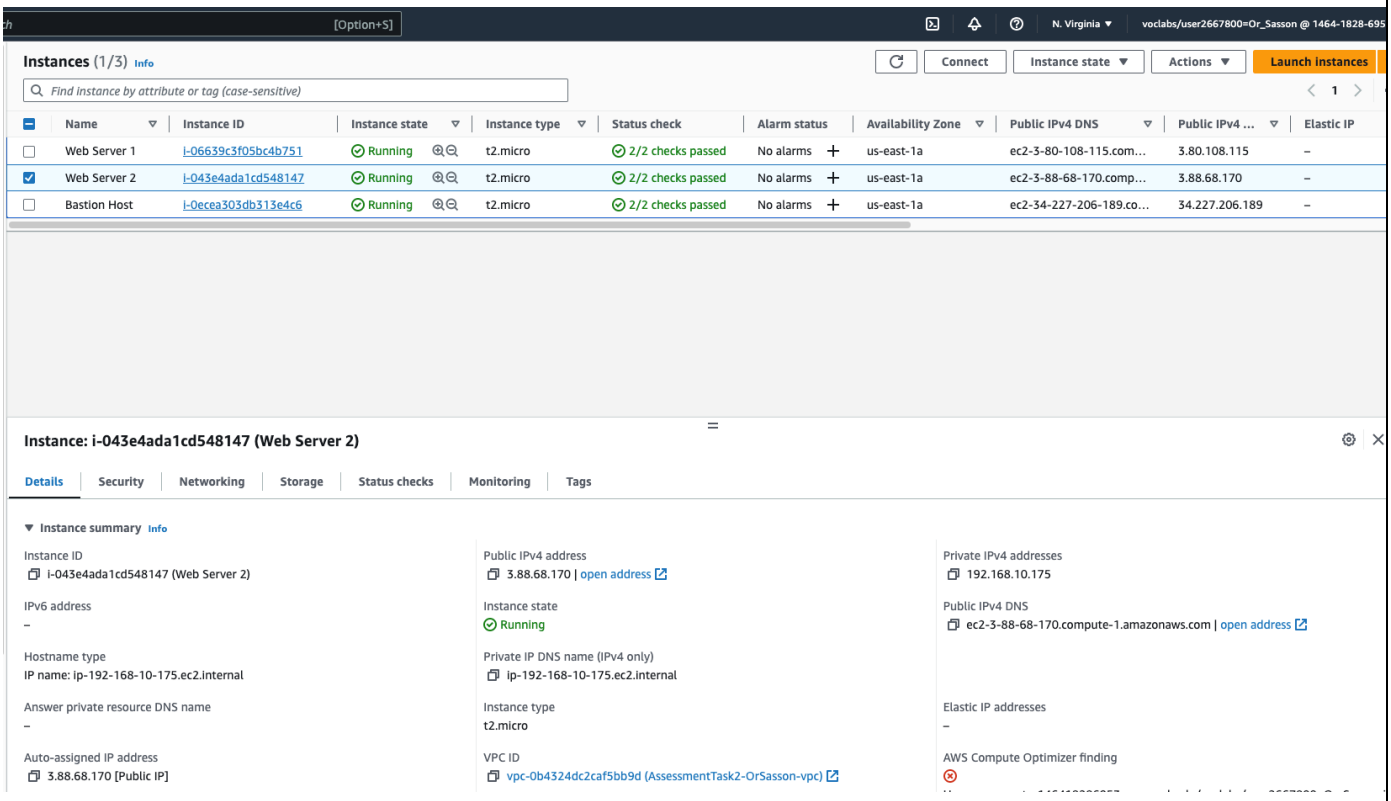
Your answer:
Provide screenshot of EC2 #1 created in subnet #1. You must include private IP and public IP in the screenshot.



For EC2 #1, verify if the private IP is correct (Yes☒ / No☐)

For EC2 #1, verify if a public IP is assigned (Yes☒ / No☐)

Provide screenshot of EC2 #1 created in subnet #2. You must include private IP and public IP in the screenshot.



For EC2 #2, verify if the private IP is correct (Yes☒ / No☐)

For EC2 #2, verify if a public IP is assigned (Yes☒ / No☐)

Part 3. Create and configure an Amazon Elastic Compute Cloud (Amazon EC2) instance

In the part, you are required to create the required Amazon EC2 virtual machine instance and host a web application

Virtual machine requirement:

- Linux operating system
- 1~2 vCPU and 1GB~4GB RAM
- 8GB~10GB hard drive

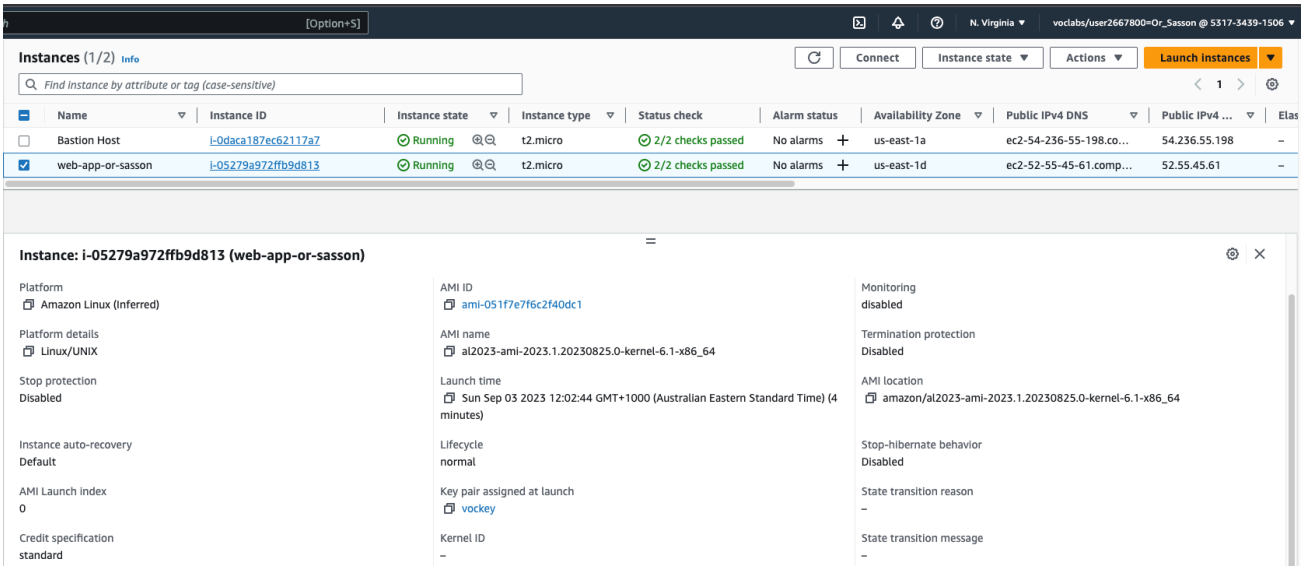
Application information:

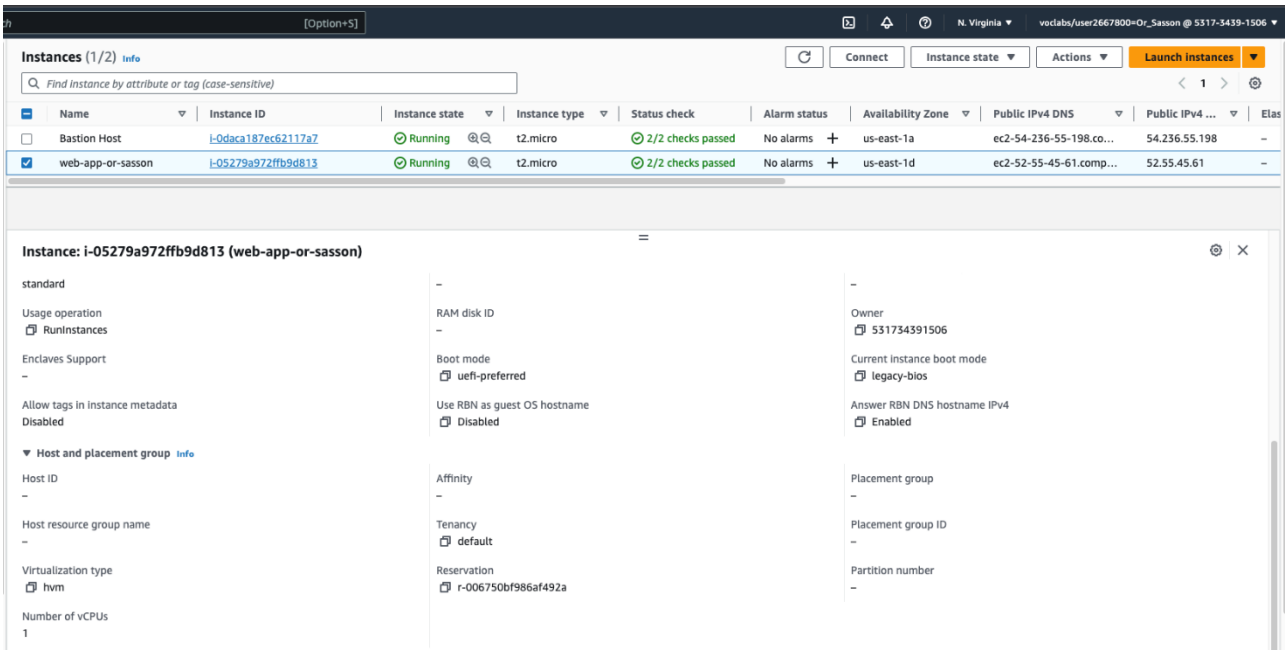
- App location: https://github.com/qiaoli116/ictcld401-python-app.git

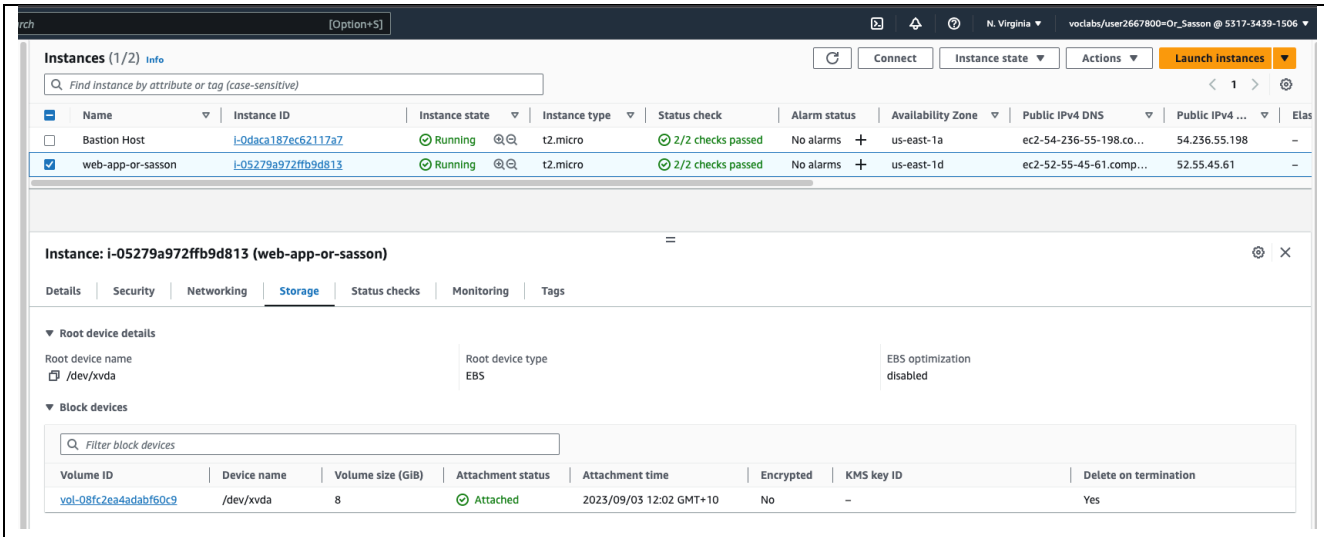
3.1. Create the required Amazon EC2 virtual machine instance.

Your answer:

Provide screenshot of EC2 instance created. In the screenshots, you must show the information of OS, CPU, memory, hard drive.



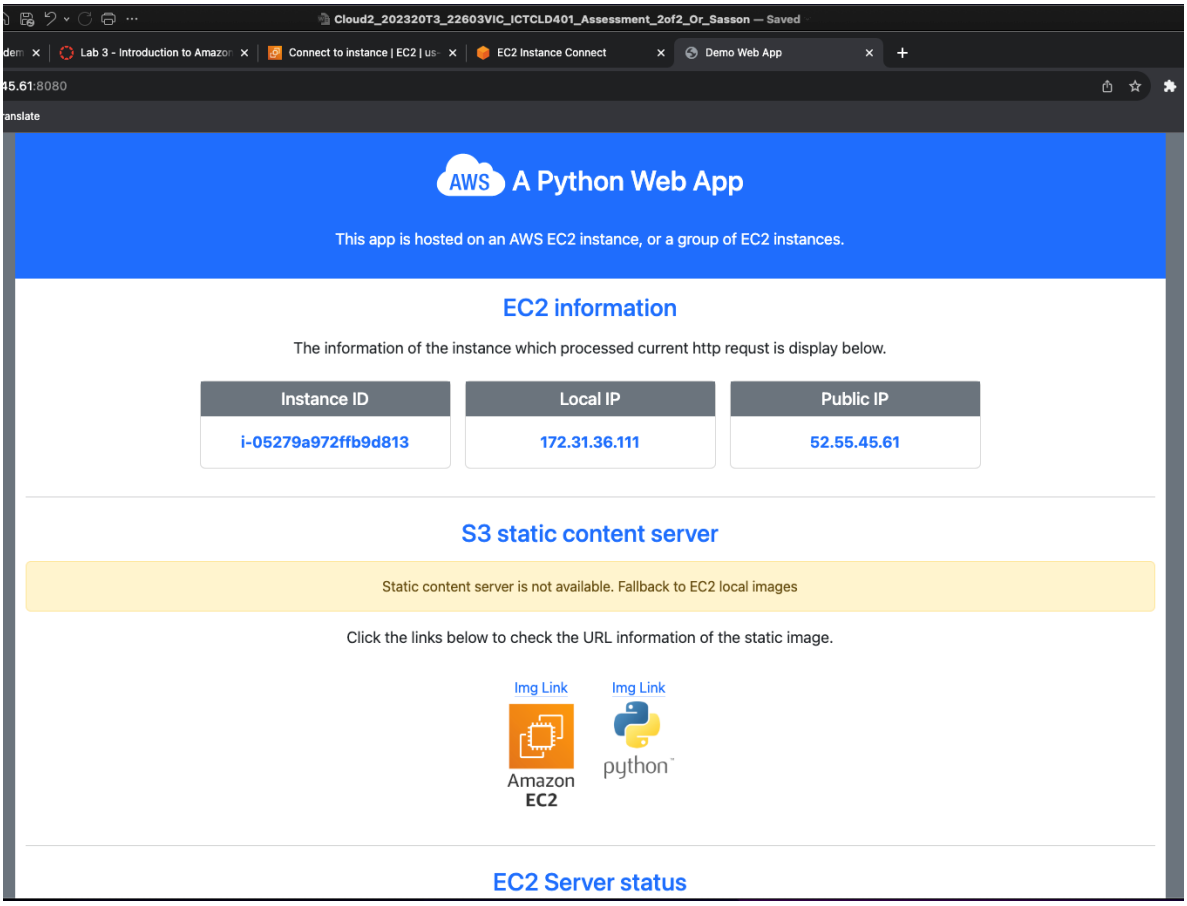




3.2. Configure the EC2 server, host the target web application and verify the result

Your answer:

Provide screenshot of to show that the application can be accessible from Internet.

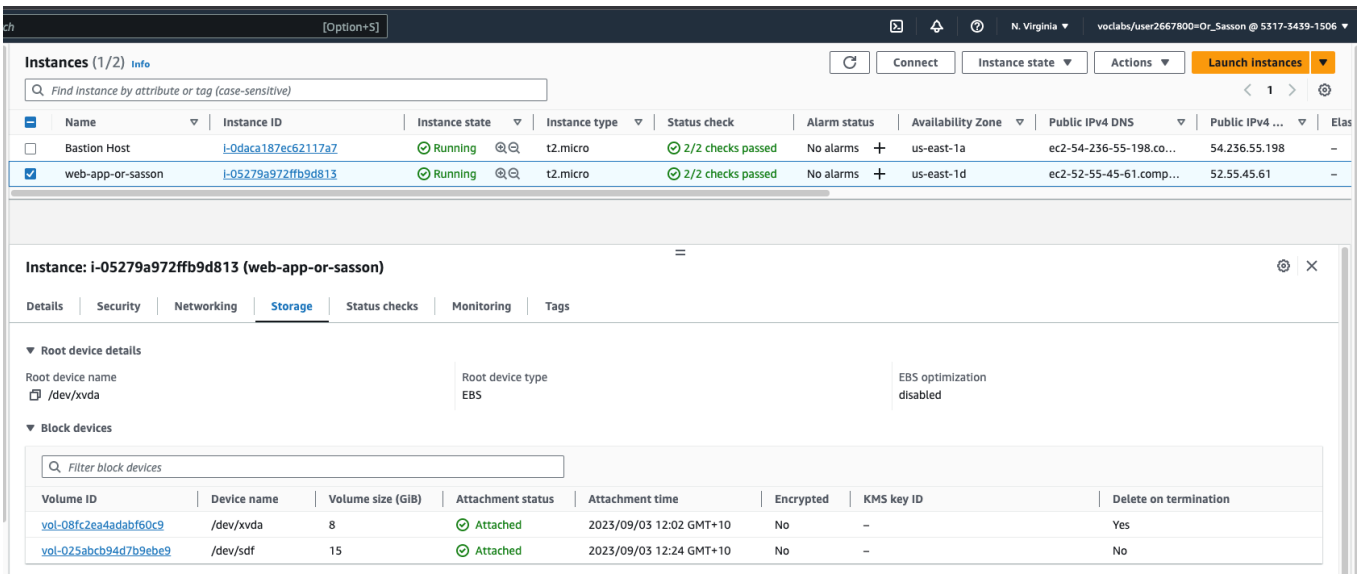


3.3. Expend the storage (hard drive) for the EC2 instance created above

Attached a second drive of 15 GB to the EC2 instance create above

Your answer:

Provide screenshot of EC2 instance created with the additional drive.



Part 4. Create and configure an Amazon Relational Database Service (Amazon RDS) instance

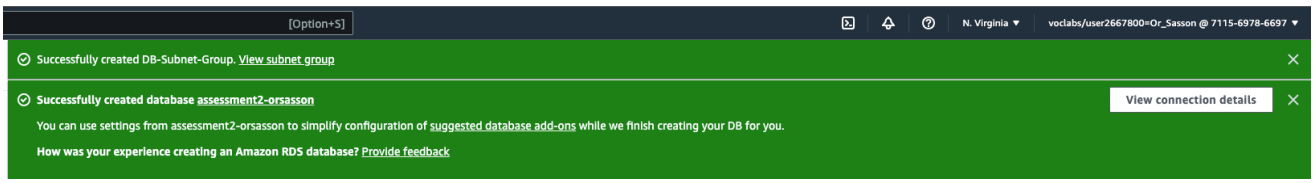
In the part, you are required to create an Amazon RDS instance with the following requirements

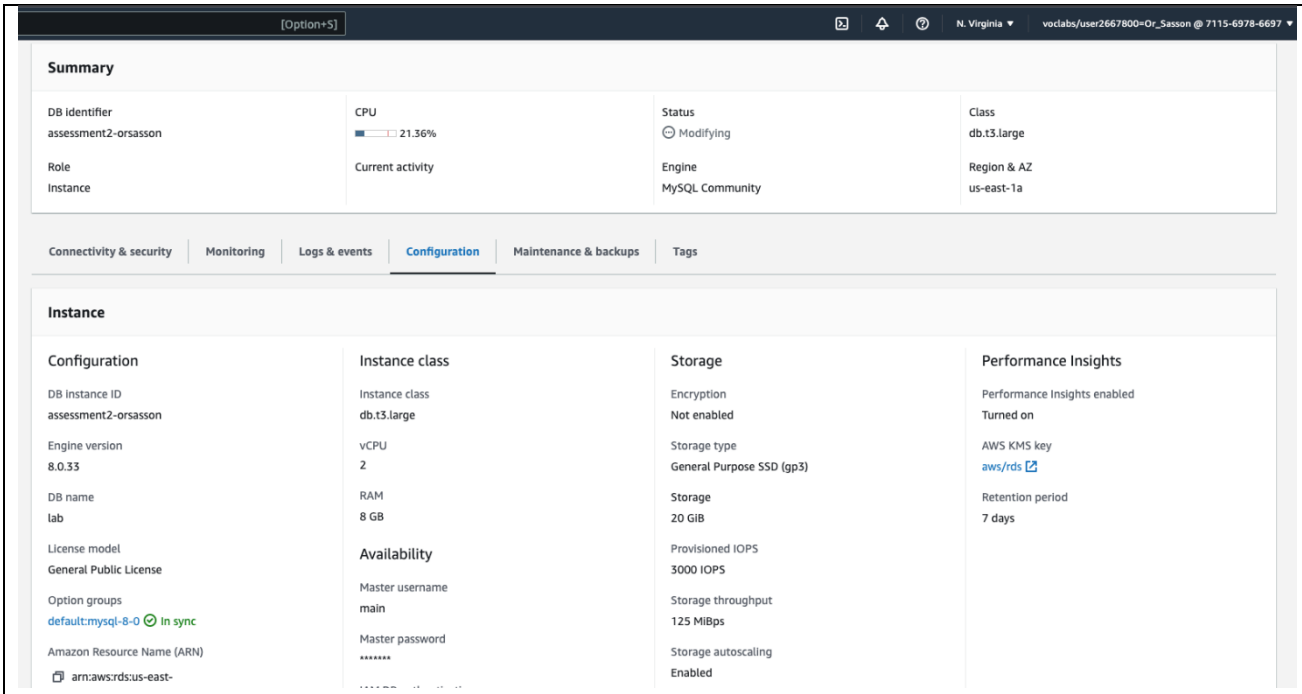
- MySQL Database
- Allow public access
- 2~4 vCPU and 8GB~16GB RAM

4.1. Create the required RDS instance

Your answer:

Provide screenshot of RDS instance created.

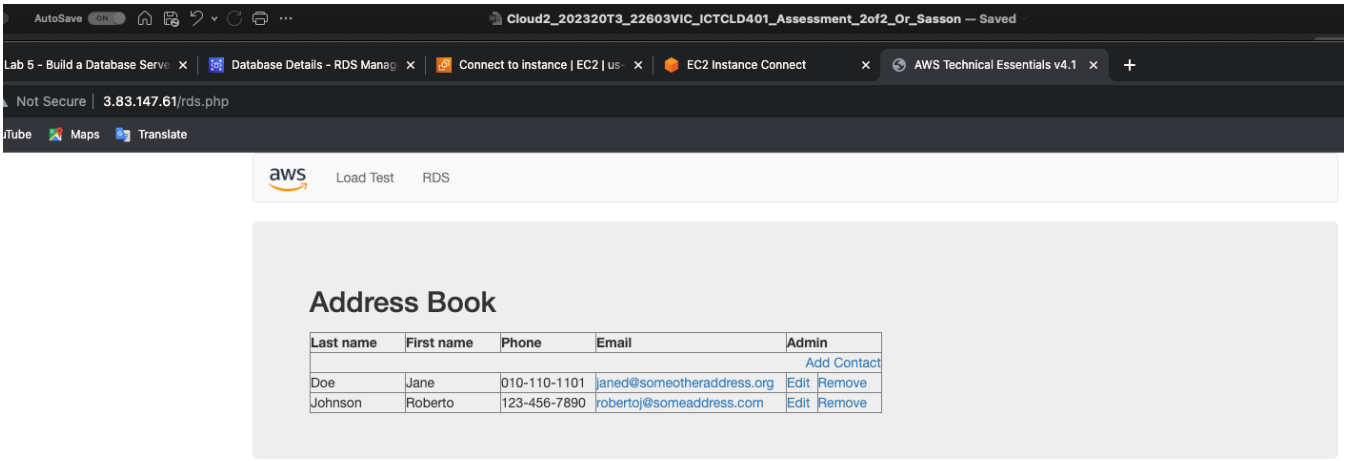




4.2. Connect to the database remotely from a local PC or an EC2 instance

Your answer:

Provide screenshot of successful remote connection to the RDS instance.



```
[ec2-user@ip-10-0-2-17 ~]$ mysql -h assessment2-orsasson.chhsmwx3nhwg.us-east-1.rds.amazonaws.com -u main -p
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MySQL connection id is 37
Server version: 8.0.33 Source distribution

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MySQL [(none)]> show databases;
+--------------------+
| Database           |
+--------------------+
| information_schema |
| lab                |
| mysql              |
| performance_schema |
| sys                |
+--------------------+
5 rows in set (0.002 sec)

MySQL [(none)]>
```

i-0767dc8ca55708c80 (Web Server 1)

PublicIPs: 3.83.147.61 PrivateIPs: 10.0.2.17

Part 5. Create a multi-layer web application

In this part, you are required to create a multi-layer web application. Here are what to be created

- Create a target group
- Create a load balancer, which connect to the target group
- Create a launch template, with user data to configure the server
- Create an auto-scaling group, which connect to the load balancer
- Configure the auto-scaling policy
- Test auto-scaling

Application information:

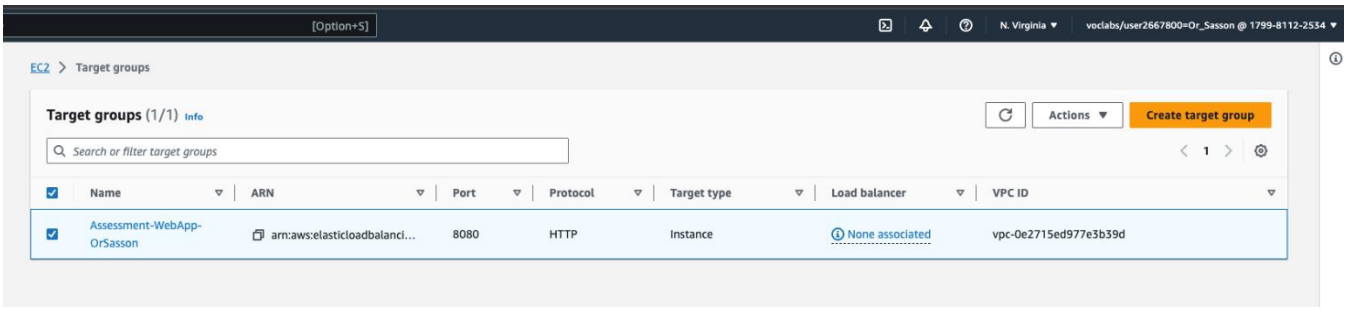
- App location: https://github.com/qiaoli116/ictcld401-python-app.git

5.1. Create and configure the required multi-layer web application infrastructure

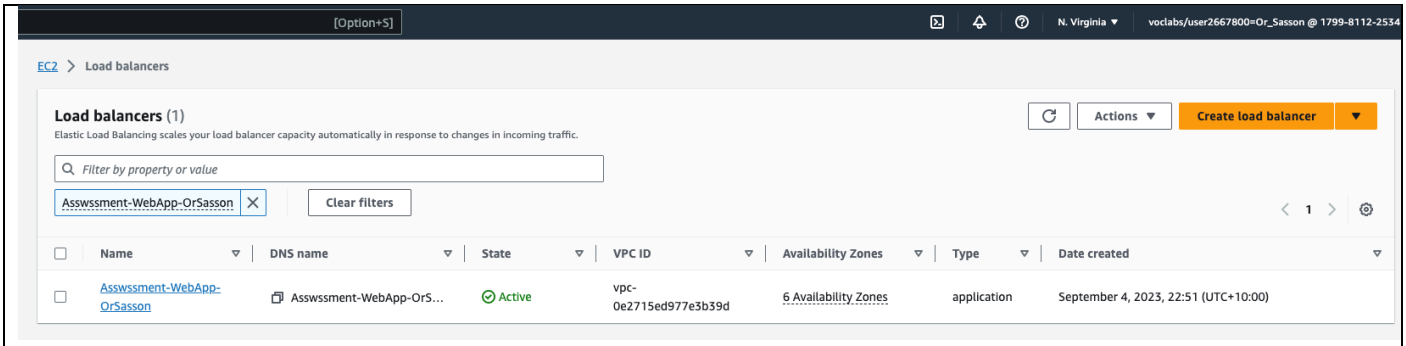
Your answer:

Provide screenshot of all the resources created. Including

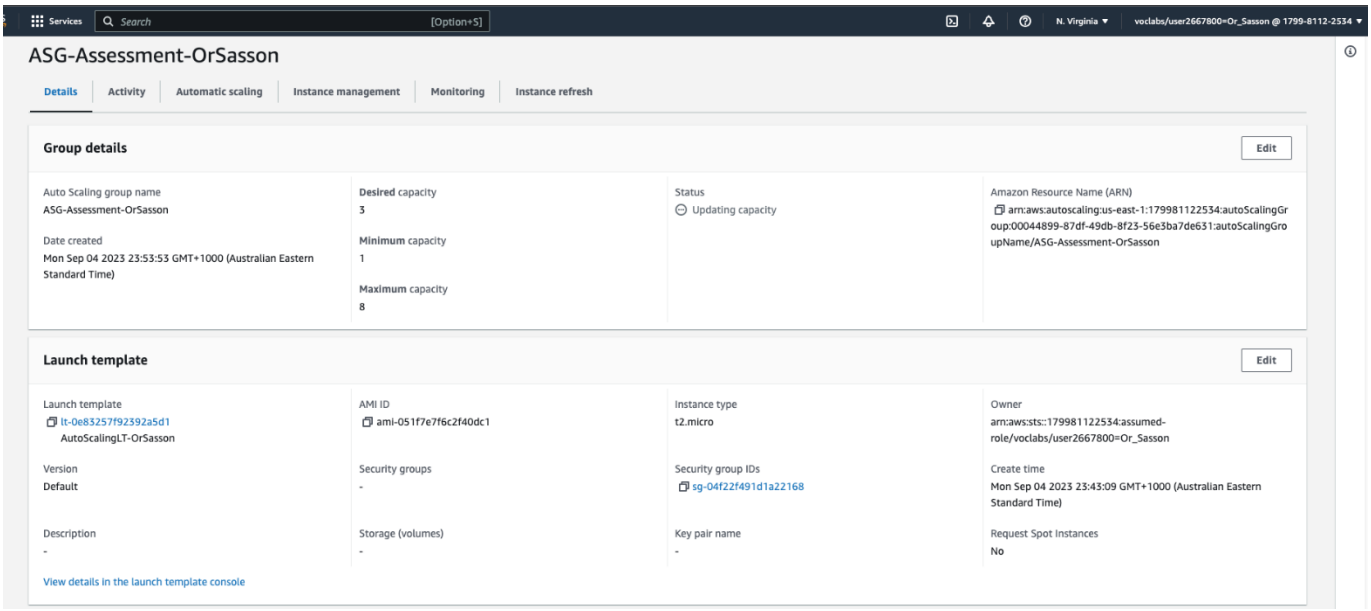
Target group:



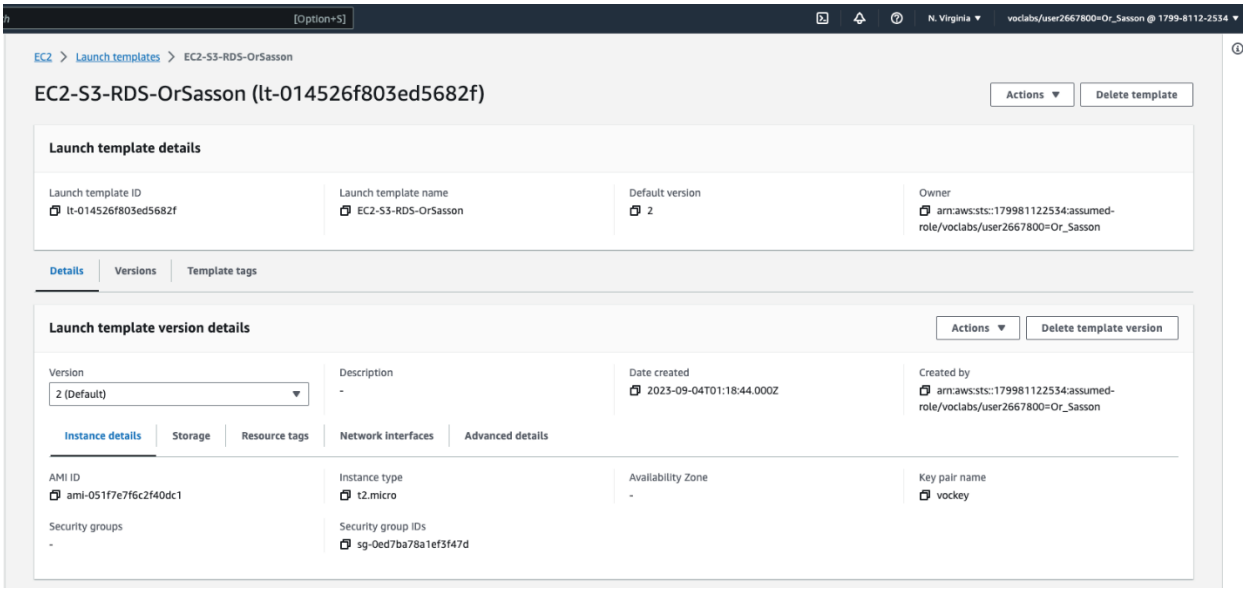
Load balancer:



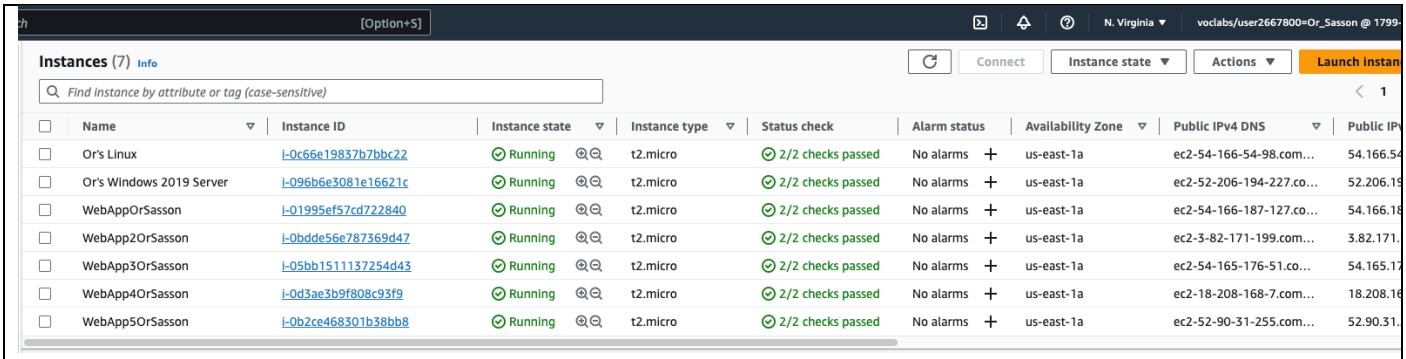
Auto scaling group:

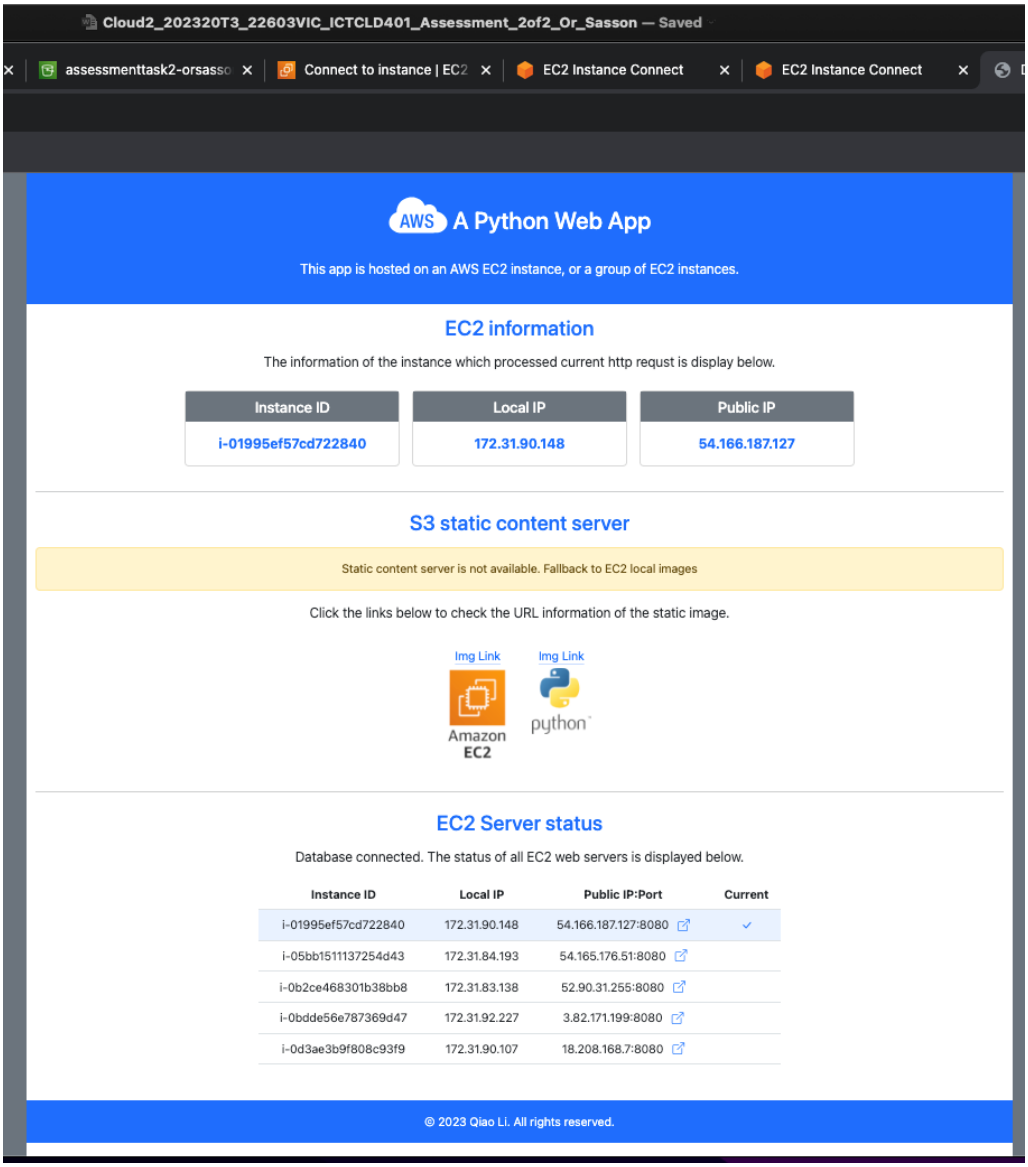


Launch template:



EC2 Instances created (at least 2 running instances):



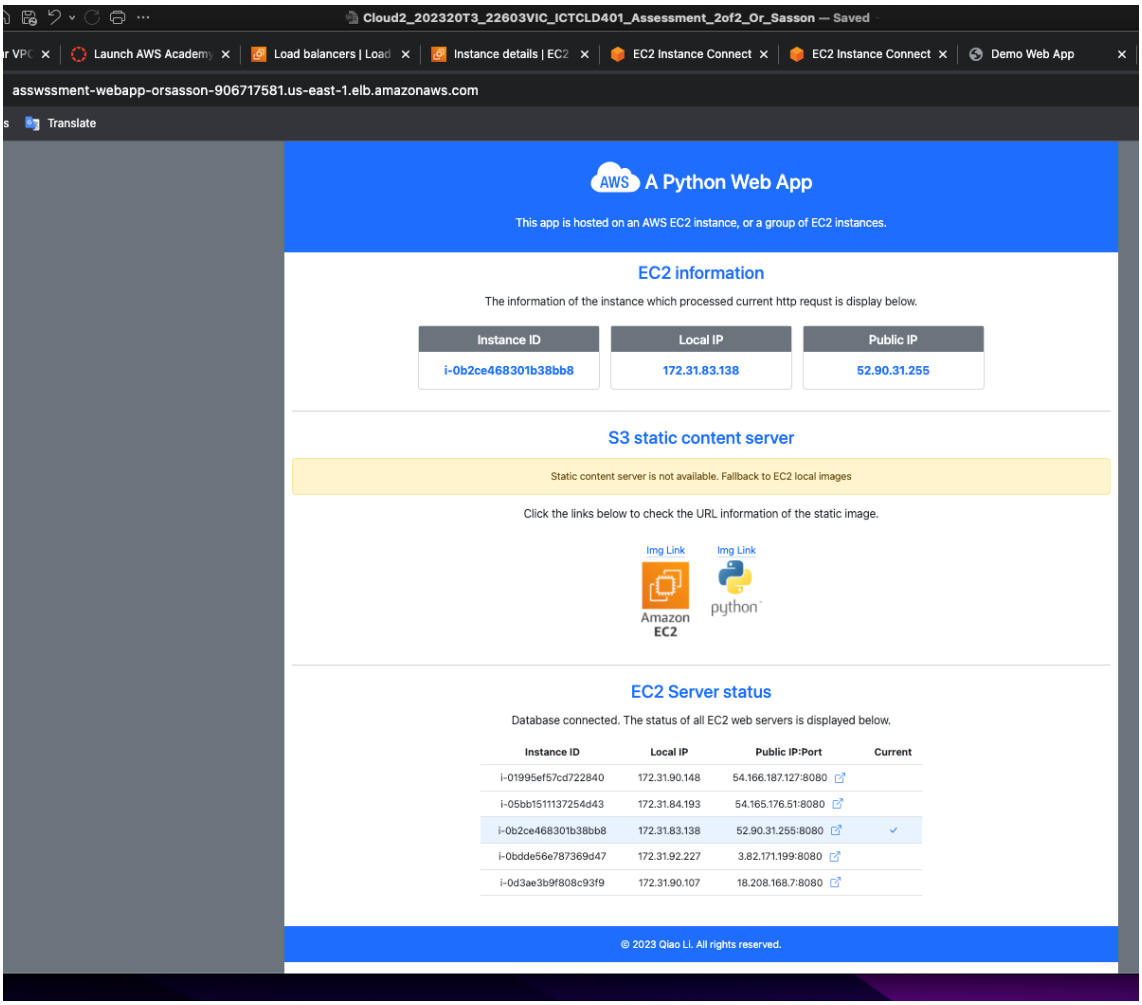


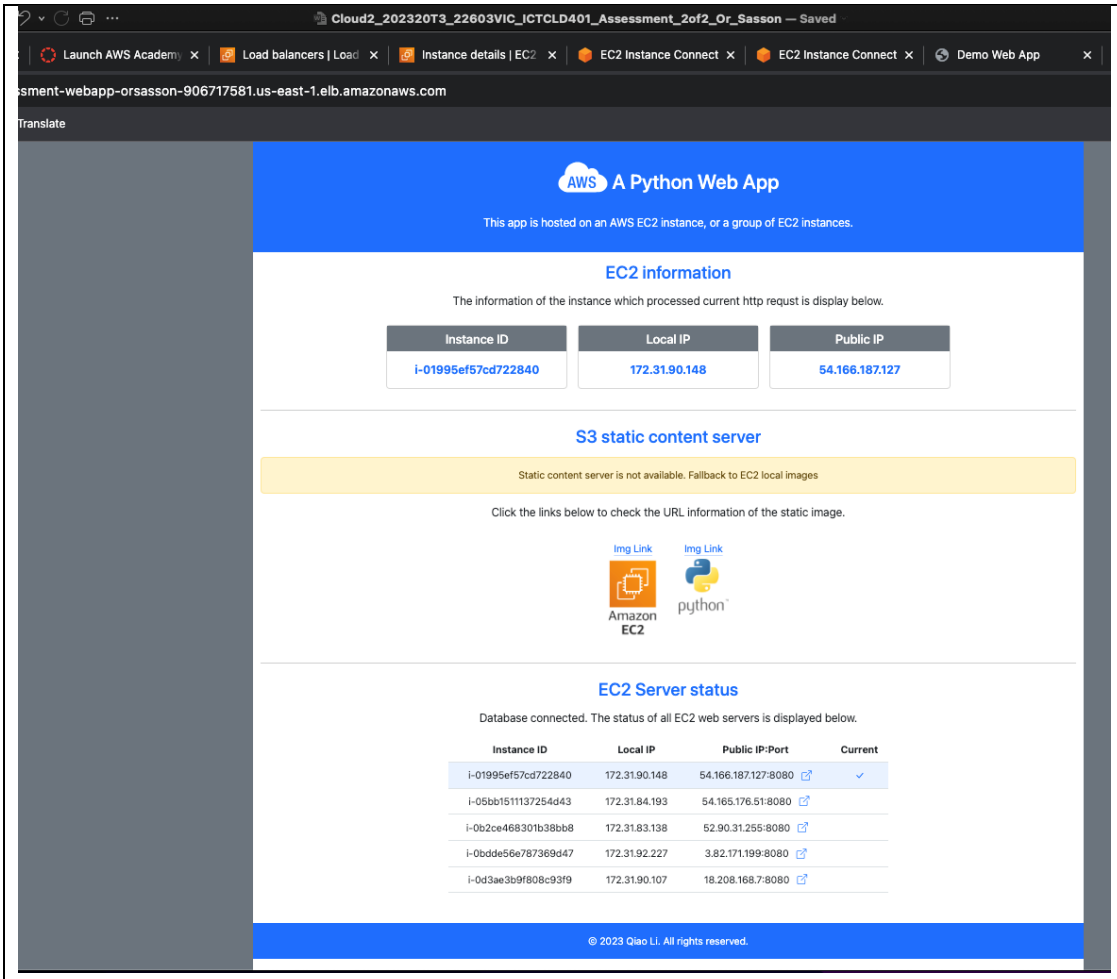
5.2. Verify the application

You must have at least two EC2 instances running behind the load balancer, in order to verify if load balancer is working as expected.

Your answer:

Provide screenshot to show that the application is up and running, and load balancer is forwarding request to different EC2 instances.


Cloud2_202320T3_22603VIC_ICTCLD401_Assessment_2of2_Or_Sasson — Saved
Launch AWS Academy
Load balancers | Load
Instance details | EC2
EC2 Instance Connect
EC2 Instance Connect
Demo Web App
asswssment-webapp-orsasson-906717581.us-east-1.elb.amazonaws.com
Translate
A Python Web App
This app is hosted on an AWS EC2 instance, or a group of EC2 instances.
EC2 information
The information of the instance which processed current http requst is display below.
Instance ID
Local IP
Public IP
i-0b2ce468301b38bb8
172.31.83.138
52.90.31.255
S3 static content server
Static content server is not available. Fallback to EC2 local images
Click the links below to check the URL information of the static image.
Img Link
Img Link
Amazon EC2
python
EC2 Server status
Database connected. The status of all EC2 web servers is displayed below.
Instance ID
Local IP
Public IP:Port
Current
i-01995ef57cd722840
172.31.90.148
54.166.187.127:8080
i-05bb1511137254d43
172.31.84.193
54.165.176.51:8080
i-0b2ce468301b38bb8
172.31.83.138
52.90.31.255:8080
i-0bdde56e787369d47
172.31.92.227
3.82.171.199:8080
i-0d3ae3b9f808c93f9
172.31.90.107
18.208.168.7:8080
© 2023 Qiao Li. All rights reserved.



5.3. Configure auto-scaling policy

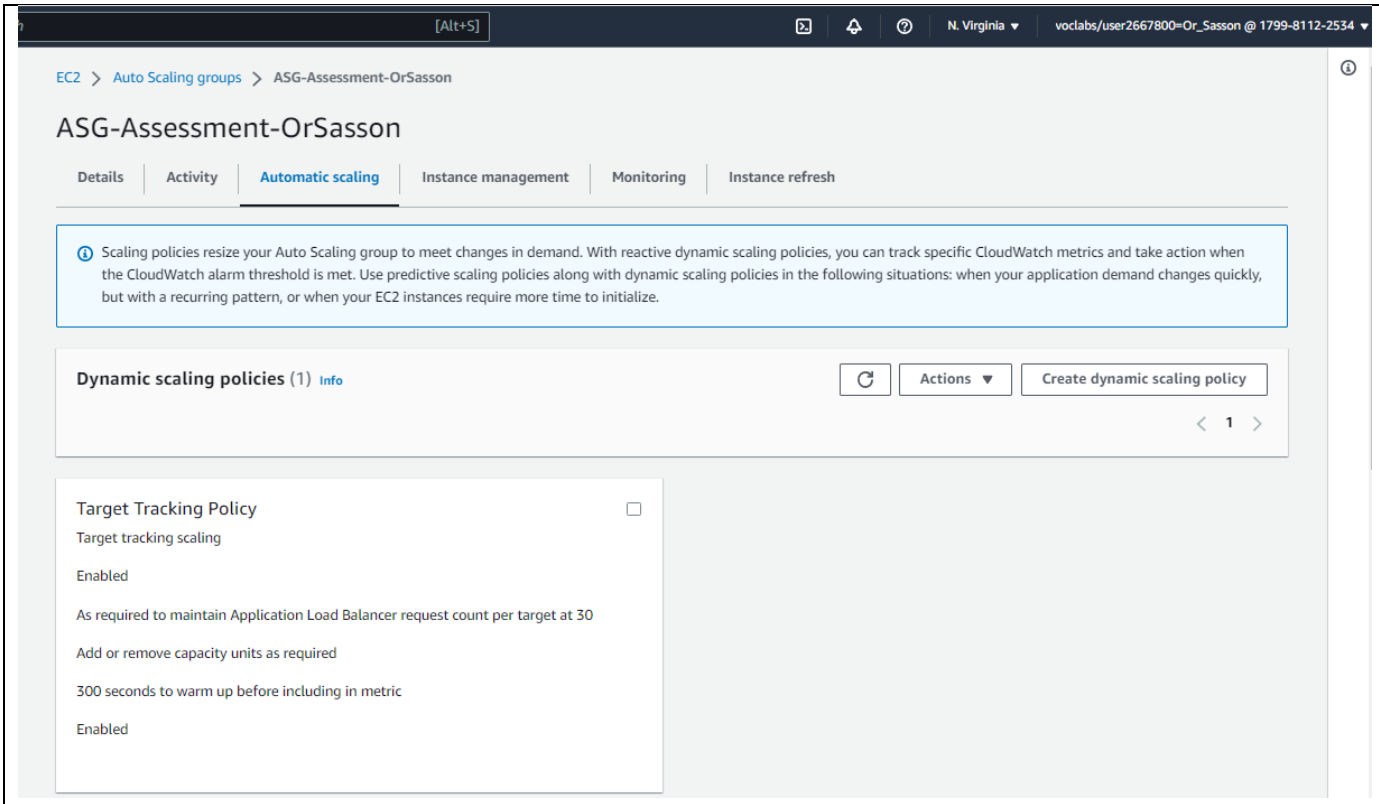
You are required to configure an auto-scaling policy, and the strategy is up to your choice.

Your answer:

Explain the auto-scaling strategy to be configured

I decided to go with predictive auto scaling policy by target metric with the ability to scale based on forecast according to the historical data, so it can predict the demand to make sure the app is always prepared for the traffic.

Provide screenshot to show that the auto-scaling policy has been configured



5.4. Test auto scaling

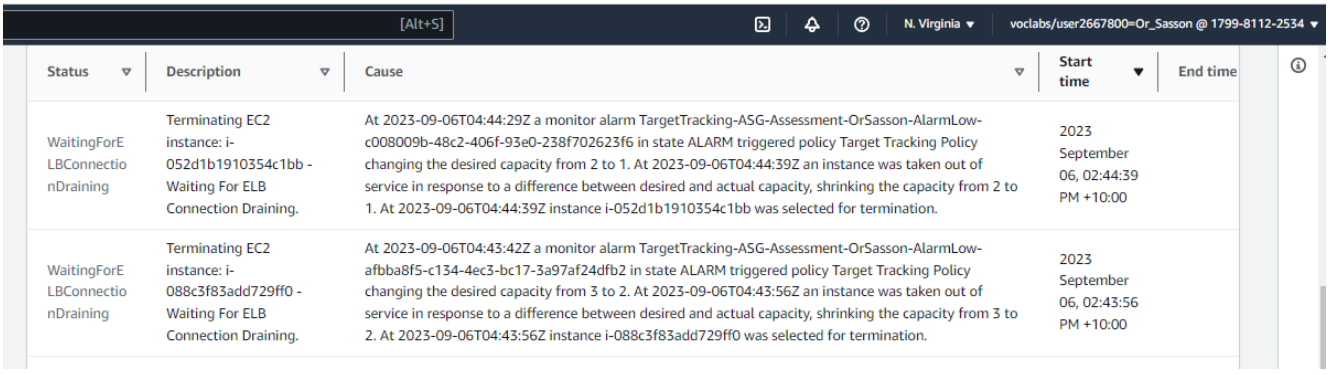
If the test failed, you must troubleshoot and fix the errors.

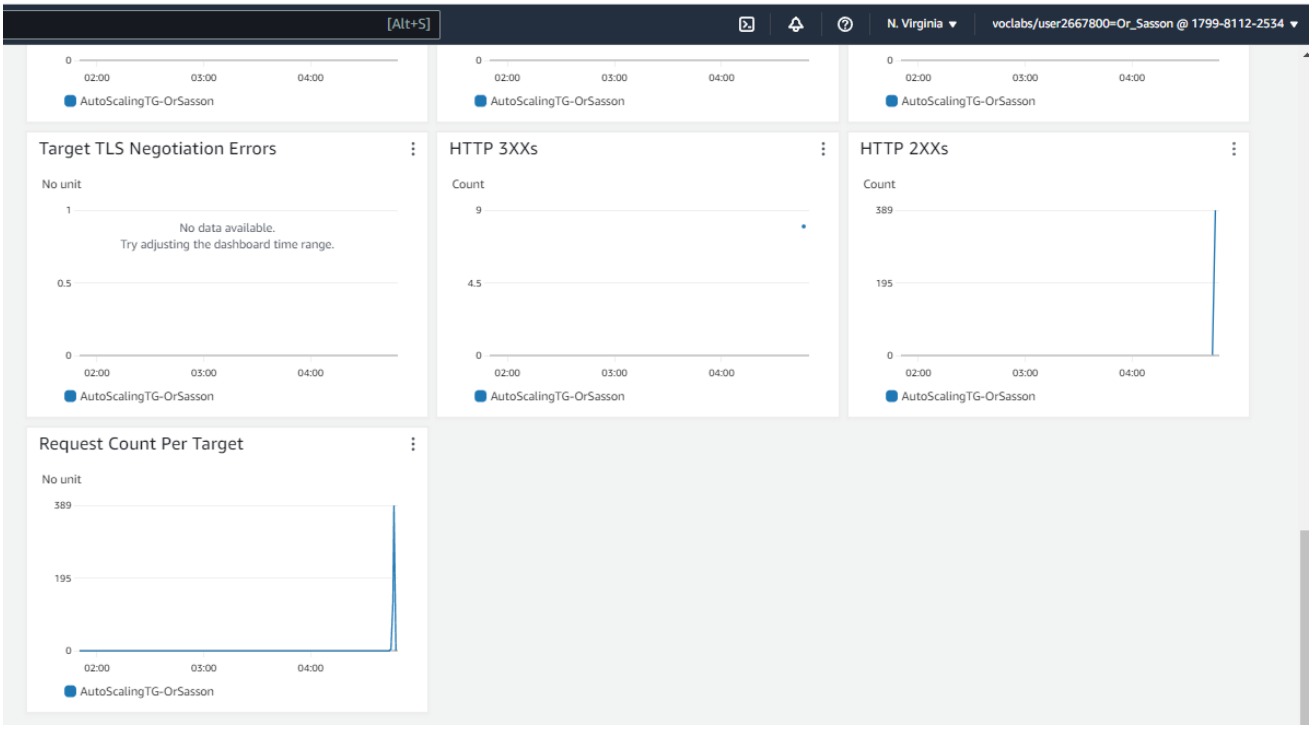
Your answer:

Explain the testing strategy according to the auto-scaling policy configured above

Using the loader.io tool to send many requests to the instance and see if the auto-scaling is increasing the numbers of used instance due to the increasing numbers of requests.

Provide screenshot / evidence to show that the auto-scaling function is working as expected





5.5. Documentation

Document the technical details of all the cloud resource created in part 5.

Your answer:

Resource	**Description and technical details**
Target group	Assessment-WebApp-OrSasson TG with port 8080 in us-east-1a zone
Load balancer	Assessment-WebApp-OrSasson Load balancer ARN: arn:aws:elasticloadbalancing:us-east-1:179981122534:loadbalancer/app/Asswssment-WebApp-OrSasson/990b3033f2264cc8 DNS name: Asswssment-WebApp-OrSasson-906717581.us-east-1.elb.amazonaws.com
Auto scaling group	ASG-Assessment-OrSasson AMI ID: ami-051f7e7f6c2f40dc1 Desired capacity: 1 Minimum capacity: 1 Maximum capacity: 8 ARN: arn:aws:autoscaling:us-east-1:179981122534:autoScalingGroup:00044899-87df-49db-8f23-56e3ba7de631:autoScalingGroupName/ASG-Assessment-OrSasson Availability zones: us-east-1a, us-east-1b, us-east-1c, us-east-1d, us-east-1e, us-east-1f
Launch template	Name: EC2-S3-RDS-OrSasson Version: 3 ID: lt-014526f803ed5682f

	Owner: arn:aws:sts::179981122534:assumed-role/voclabs/user2667800=Or_Sasson Instance type: t2.micro Key pair name: vockey
auto-scaling policy	Metric pair: ALB request count 30 requests received per instance Target utilization: 30 Forecast and scale: enable

5.6. Seek and respond to feedback

Document and communicate work to client or supervisor, Seek respond to their feedbacks

Your answer:

Feedback 1:

Everything was configured correctly, except the policy which was by the CPU metric and not by the target.

Actions:

I changed my answer in Q5.3 for a target metric with the relevant screenshot.

Feedback 2:

Everything is working perfectly.